Proposed Algorithm: Explanation Guided Ensemble Selection (XGES)

Input:

Dataset D

Model set M

Sampling size s

Number of selected models k

Output:

Optimized stacked ensemble E

Step 1: Data Preparation

Load dataset D

Split dataset into Train and Test using stratified sampling

Apply SMOTE to training data

Scale features using standardization

Step 2: Train Candidate Models

For each model m in M do

Train m using balanced training data

Store trained model

End For

Step 3: Performance Evaluation

For each trained model m do

Predict probabilities on test set

    Compute AUPRC score

    Save performance score

  End For

Step 4: Create Stratified Subset

  Sample subset S from training data

  Maintain class distribution

Step 5: Generate Explanations

  For each model m do

    Apply SHAP on subset S

    Extract feature importance vector

  End For

Step 6: Compute Stability

  For each model m do

    Perform k-fold analysis on subset S

    Measure variance of feature importance

    Compute stability score

  End For

Step 7: Compute Diversity

  For each pair of models (mi, mj) do

    Compare explanation vectors

    Calculate similarity score

    Convert to diversity score

  End For

Step 8: Model Ranking

For each model m do

Score(m) =

$\alpha \times$ Performance

$+ \beta \times$ Stability

$+ \gamma \times$ Diversity

End For

Step 9: Model Selection

Rank models by Score

Select top-k models

Step 10: Stacking

Train selected base models

Generate predictions

Train meta-learner using predictions

Step 11: Final Evaluation

Evaluate ensemble on test data

Return Ensemble E